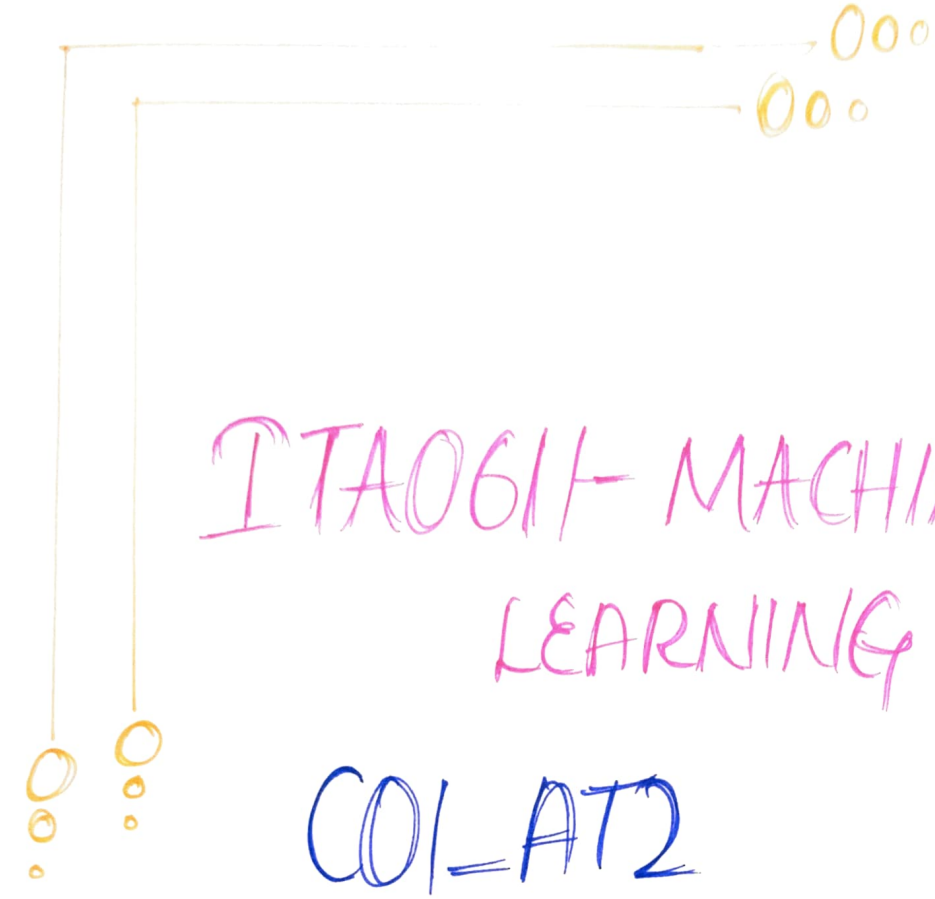


ITA0611- MACHINE LEARNING

CO1_AT2

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MACHINE LEARNING SYSTEM - BLOCKS & LEARNING METHODS

1.

MACHINE LEARNING SYSTEM

Data
- Raw data
- Sources: files, databases, sensors, Web, etc

Preprocessing
- cleaning
- feature scaling
- Feature selection

Learning Algorithm
- supervised
- unsupervised
- Reinforcement

Model
- training model
- Learns patterns from data

Evaluation
- Accuracy
- F1-score
- Cross validation

Deployment
- use in real world
- Monitor and Update.

EXAMPLES OF LEARNING METHODS

SUPERVISED LEARNING

Uses labelled data (input + output)

Eg: E-mail classification (Spam / Not Spam)

UNSUPERVISED LEARNING

Uses unlabelled data (finds patterns)

Eg: Customer segmentation (Grouping similar customers)

LEARN

2. DISASTER MANAGEMENT SYSTEM - FLOODS & EARTHQUAKES

DISASTER MANAGEMENT ML SYSTEM

DATA SOURCES
- Historical data
- Real-time data

Eg: Rainfall, seismic, weather

DATA PREPROCESSING
- Handle missing values
- Remove noise
- convert to useful format

FEATURE SELECTION
- Relevant features

Eg: Rainfall, soil, seismic activity
- Methods: Filter, Wrapper, Embedded

MODEL CHOICE
- Time-series: LSTM
- classification Random-forest / XGBoost
- Good for complex patterns & accuracy

TRAINING → VALIDATION → EVALUATION

(split data, true parameters, test on unseen data)

EARLY WARNING
→ Predicts risk levels

→ Alerts authorities & public.

→ Issues alerts

EMERGENCY RESPONSE
→ Helps in evacuation planning

→ Allocates resources

→ Reduces loss of life and property

FIND-S ALGORITHM (TENNIS DATASET)

DATASET

(outlook, temp
Erature, Wind
→ play
tennis)

FIND S-ALGORITHM

Finds the most specific hypothesis consistent with positive examples

STEPS

1. Start with most specific hypothesis
2. For each +ve example, compare & generalize the hypothesis
3. Update only if attribute value differs

RESULT

Maximally specific hypothesis

Example (Tennis dataset) Update,

Positive example	outlook	Temp	Humidity	Wind	hypothesis
Initial	∅	∅	∅	∅	<?,?,?,?>
Ex 3(Yes)	Overcast	Hot	High	Weak	<Overcast, Hot, High, Weak>
Ex 4(Yes)	Rainy	Mild	High	Weak	<Rainy, Mild, High, Weak>
Ex 5(Yes)	Rainy	Cool	Normal	Weak	<Rainy, Cool, Normal, Weak>

: 4.

CANDIDATE ELIMINATION ALGORITHM (SPORT DATASET)

Candidate Elimination Algorithm

Finds the version space using two boundaries

* S * G

S - Specific Boundary

Most specific hypothesis

Initially: <ϕ, ϕ, ϕ, ϕ>

G - General Boundary

Most general hypothesis

Initially: <?, ?, ?, ?>

Procedure

→ For each training example

a. If +ve: generalizes S, remove from G

b. If -ve: specialize G, remove from S

→ Update version space after each example

Version space

(All hypotheses blw S & G)

D- S ALGORITHM (CUSTOMER BUYS PRODUCT)

FIND S-ALGORITHM

(Maximally specific hypothesis)

↓

DATASET ATTRIBUTES

Age, Income, Student, Credit rating

(use only positive examples)

↓

PROCESS

For each positive example

- Compare with current hypothesis
- Replace inconsistent values with specific values.
- Continue until all positive examples are processed

↓

STEP-by-STEP UPDATES

Init
(?,?,?,?)

Ex 3
(Middle, High, No, fair)

Ex 4
(Senior, Medium, No, fair)

↓

FINAL HYPOTHESIS

(Senior, Medium, No, fair)